

FORECASTED STREAMFLOW BULLETIN

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WATER RESOURCES AUTHORITY



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PROBLEM

- Limited Early Warning Advisories to the General Public regarding changing streamflow levels
- Population:** Approximately **40% of the population** affected
- Location:** Flood-prone Areas across Jamaica

GOAL: To provide sufficient, frequent and reliable early warning advisories to the general public about streamflow levels

AREA OF STUDY

The community of Bog Walk, St. Catherine (X: -77.0054 Y:18.098292). The Rio Cobre and tributaries including Rio Doro and Thomas River run through this area.

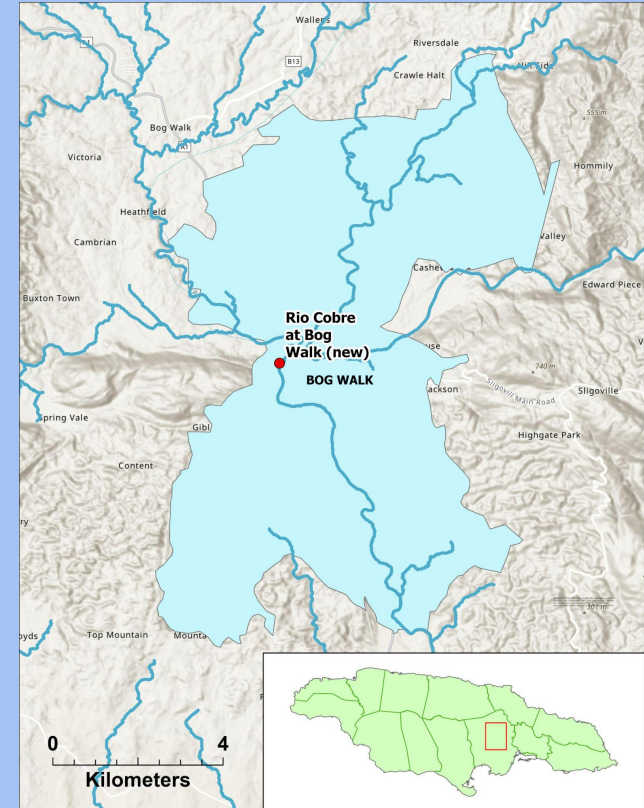
Estimated Population: 15, 389 (Social Development Commission)

Land Use:

- Agriculture
- Residential
- Industrial
- Commercial



MAP LOCATION



ASSUMPTIONS

- All persons in the area would be affected by flooding
- All historical data was verified and is accurate
- Flooding is caused by the river system in the area

CONTEXT

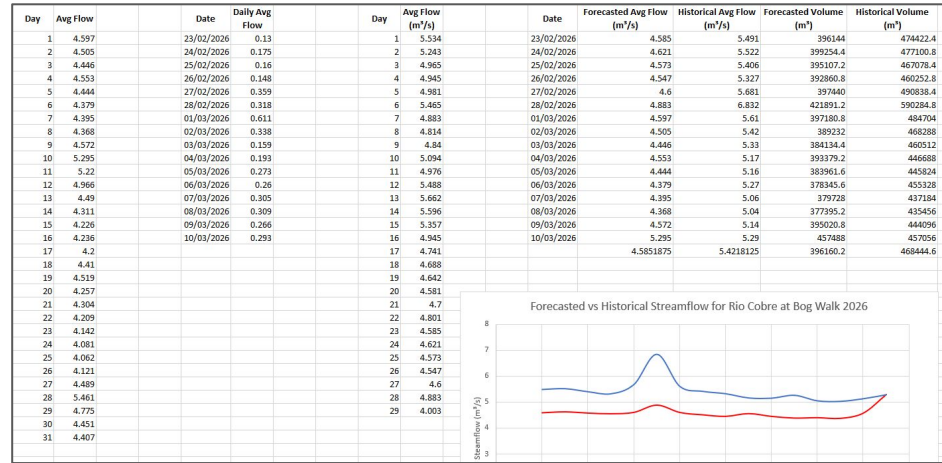
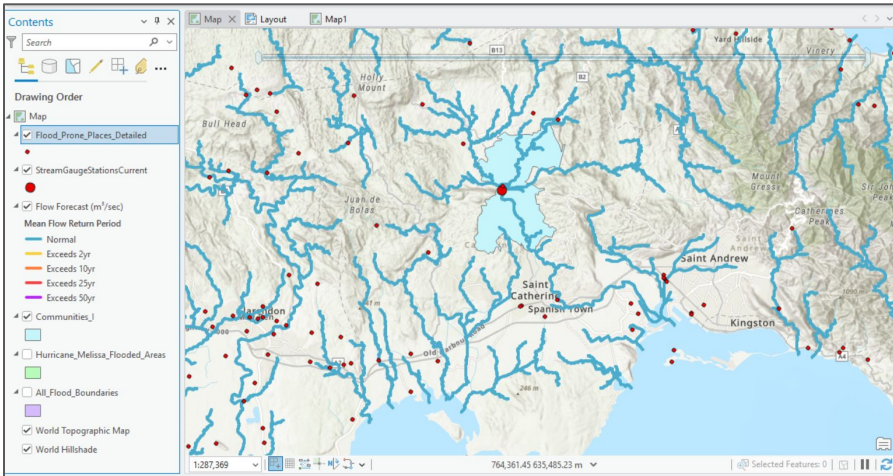
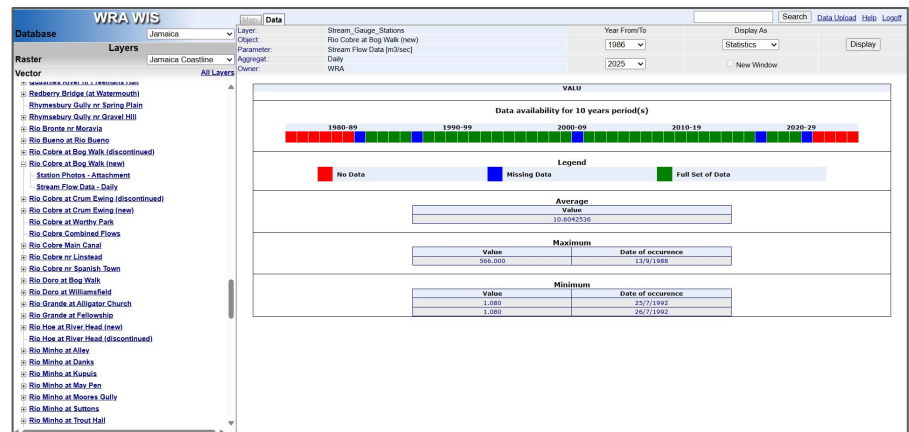
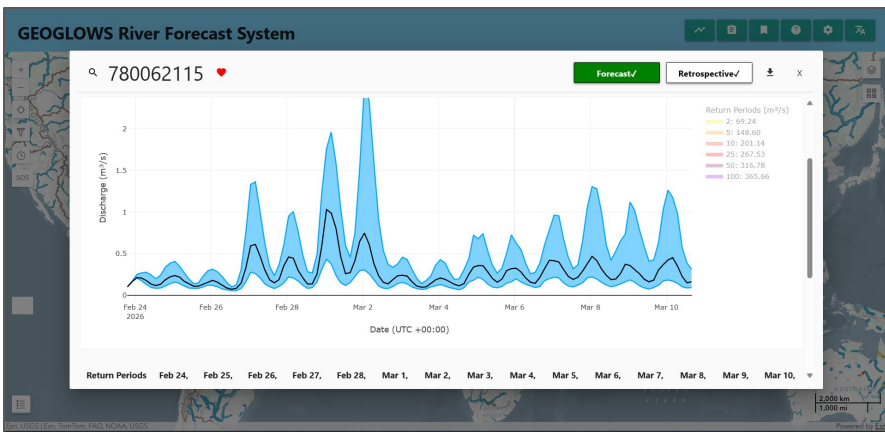
Who: Persons in Flood Prone areas are most affected by the limited early warning systems and bulletins

What: The limited early warning systems and bulletins can affect the preparedness and awareness of the populations of these flood prone communities to flooding risks. There are several reports as well as studies conducted on flooding and flood impacts in Jamaica, showing the need for sufficient and accessible data to warn vulnerable communities.

When: With an increased population, more persons live in vulnerable areas, increasing the number of persons at risk of experiencing negative flood impacts.

DATA/APPLICATIONS USED

1. **ArcGIS Pro** to import Stream Gauge Stations and GeoGLOWS through Living Atlas (identify flood locations and river ID)
2. **Hydroviewer** to view Forecasted Chart and download corresponding CSV
3. **WRA's Water Information System** to download historical streamflow data for Rio Cobre at Bog Walk (new)
4. Analyzed 10-year historical streamflow vs forecasted streamflow for Feb 23 - March 10 using **Microsoft Excel** and **AI**
5. Calculated **volume** ($\text{m}^3/\text{s} \times 86400$) for each forecasted and historical data for analysis
6. Created **line chart** of forecasted vs historical daily streamflow average
7. Created **sample streamflow bulletin** that would be made available on WRA's website and the more severe stations posted on social media



RESULTS OR EXPECTED RESULTS



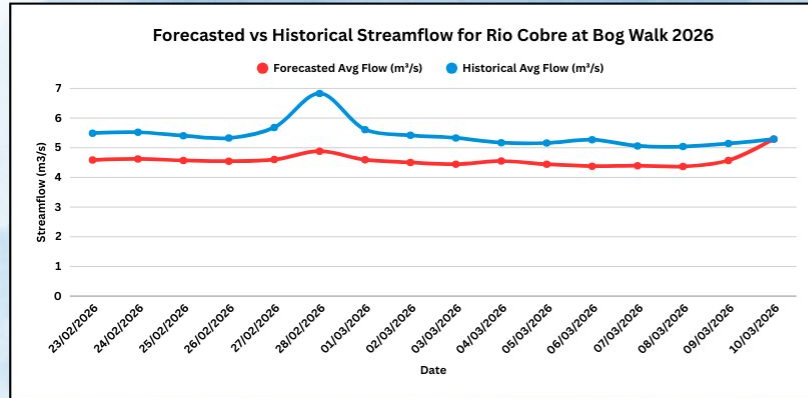
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GOGLOWS
GLOBAL WATER SUSTAINABILITY

RIVER VOLUME ADVISORY

FEBRUARY 23 - MARCH 10 2026

Stream Gauge Station:
RIO COBRE AT BOG WALK (NEW)



4.6 (m³/s)



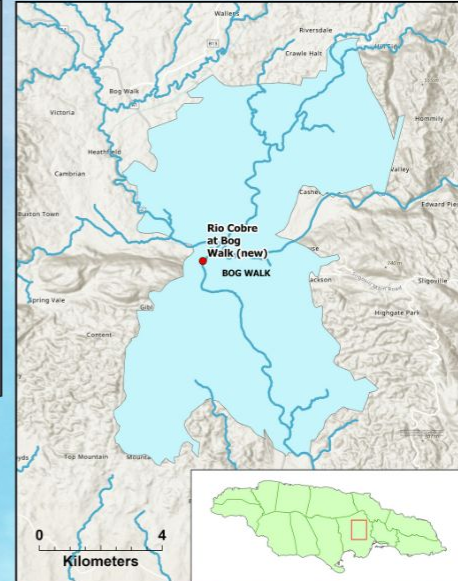
Forecast Total Volume:

6,338,563 m³

Historical Total Volume:

7,495,114 m³

15.4%



Produced by the Water Resources Authority 2026
For more information: info@wra.gov.jm
Visit our website: wra.gov.jm

Conclusions

Why was it useful or how can the data/applications be used in the future?

The outputs of this project would provide bulletins that would include forecasts, to be able to advise the public on any impending risks and help to improve public safety and water availability. With the ability to compare the historical data, and use bias correction to improve data obtained.

Recommendations

This can be expanded to all stream gauge stations in high risk flood prone areas. Furthermore, the aim is to have bulletins for all stations, with focus on those with greater variations and impending risks, both droughts and floods.

When you have the presentation ready, please upload it to the drive.

Jamaica:

https://drive.google.com/drive/folders/1pMdyM_WnN_rTqUm-hr194AMHP9EMQOLp?usp=sharing